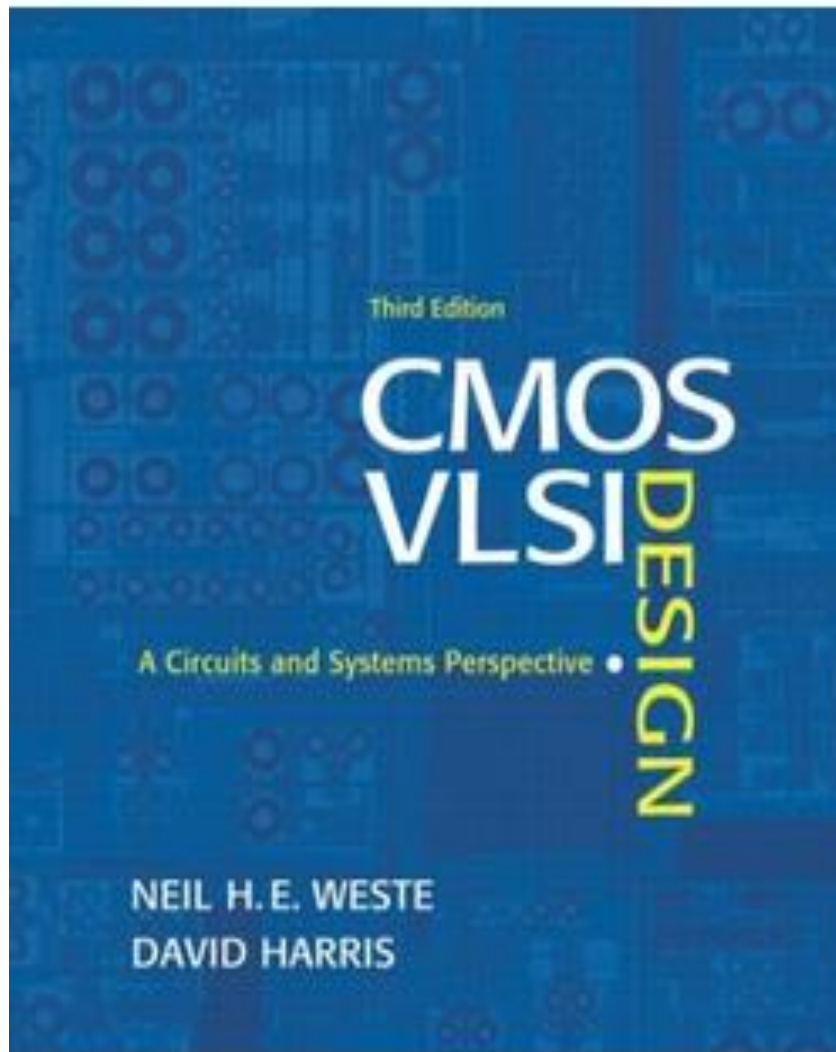




Lecture 7: Power

Outline

- ☐ Power and Energy
- ☐ Dynamic Power
- ☐ Static Power

Power and Energy

- ❑ Power is drawn from a voltage source attached to the V_{DD} pin(s) of a chip.
- ❑ Instantaneous Power: $P(t) =$
- ❑ Energy: $E =$
- ❑ Average Power: $P_{avg} =$

Power in Circuit Elements

$$P_{VDD}(t) = I_{DD}(t)V_{DD}$$



$$P_R(t) = \frac{V_R^2(t)}{R} = I_R^2(t)R$$



$$\begin{aligned} E_C &= \int_0^{\infty} I(t)V(t)dt = \int_0^{\infty} C \frac{dV}{dt} V(t)dt \\ &= C \int_0^{V_C} V(t)dV = \frac{1}{2} CV_C^2 \end{aligned}$$



Charging a Capacitor

- When the gate output rises
 - Energy stored in capacitor is
- But energy drawn from the supply is

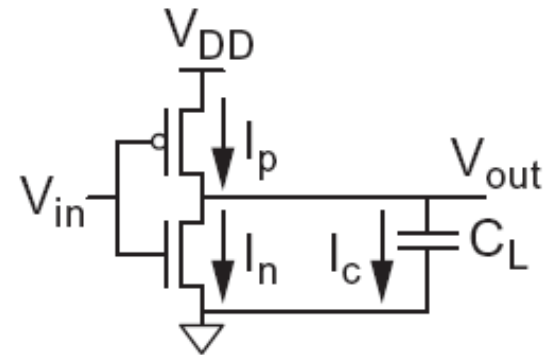
$$E_C = \frac{1}{2} C_L V_{DD}^2$$

$$E_{VDD} = \int_0^\infty I(t) V_{DD} dt = \int_0^\infty C_L \frac{dV}{dt} V_{DD} dt$$

$$= C_L V_{DD} \int_0^{V_{DD}} dV = C_L V_{DD}^2$$

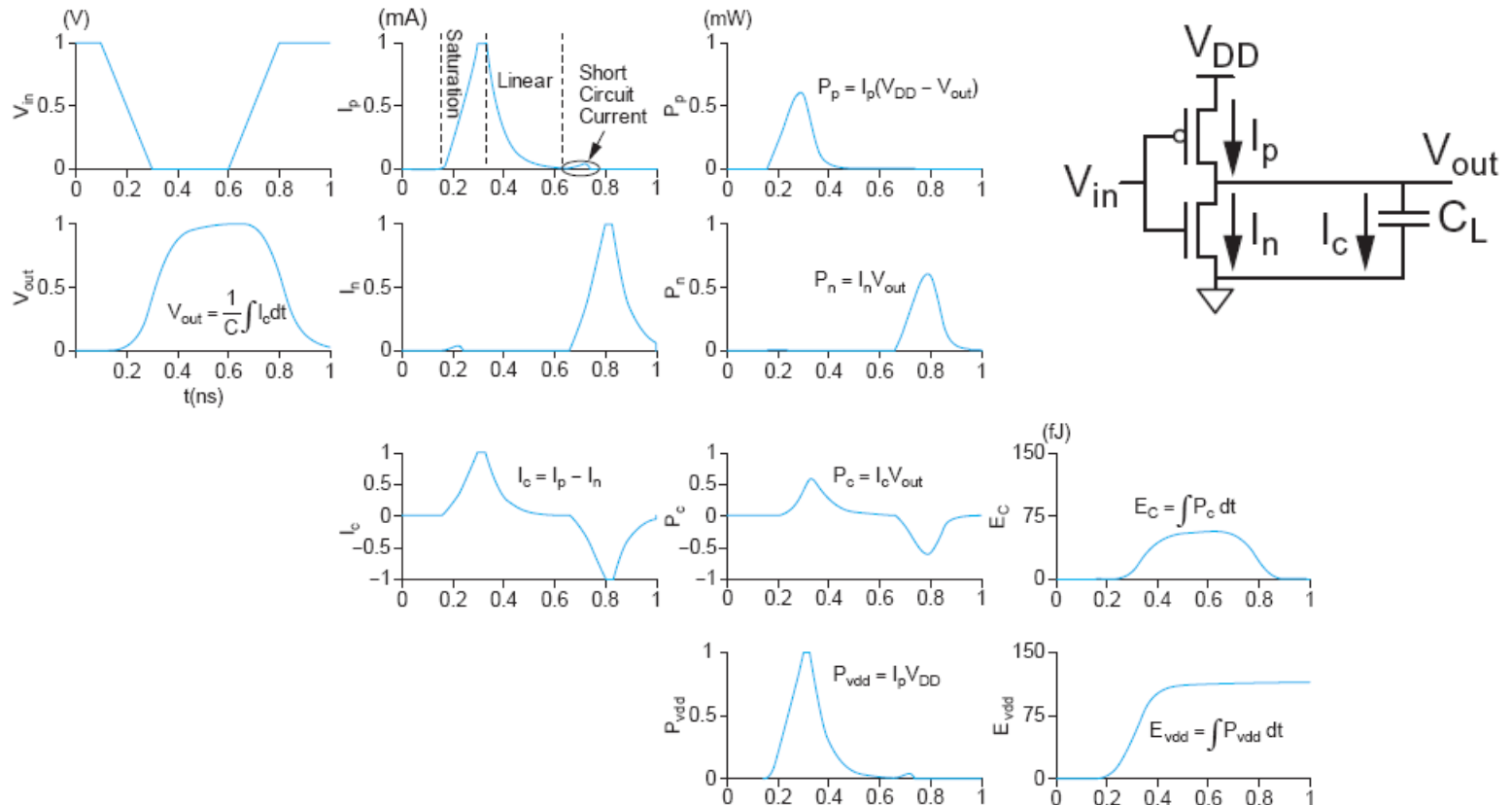
- Half the energy from V_{DD} is dissipated in the pMOS transistor as heat, other half stored in capacitor

- When the gate output falls
 - Energy in capacitor is dumped to GND
 - Dissipated as heat in the nMOS transistor



Switching Waveforms

□ Example: $V_{DD} = 1.0 \text{ V}$, $C_L = 150 \text{ fF}$, $f = 1 \text{ GHz}$



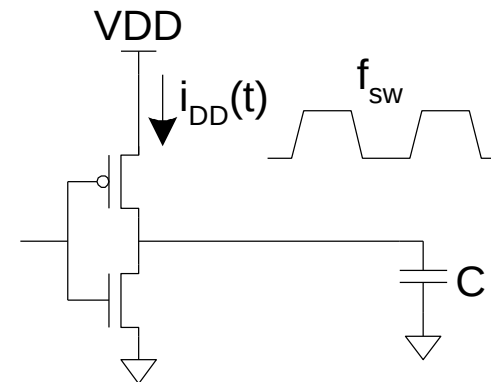
Switching Power

$$P_{\text{switching}} = \frac{1}{T} \int_0^T i_{DD}(t) V_{DD} dt$$

$$= \frac{V_{DD}}{T} \int_0^T i_{DD}(t) dt$$

$$= \frac{V_{DD}}{T} [T f_{\text{sw}} C V_{DD}]$$

$$= C V_{DD}^2 f_{\text{sw}}$$



Activity Factor

- ❑ Suppose the system clock frequency = f
- ❑ Let $f_{sw} = \alpha f$, where α = activity factor
 - If the signal is a clock, $\alpha = 1$
 - If the signal switches once per cycle, $\alpha = 1/2$

- ❑ Dynamic power:

$$P_{\text{switching}} = \alpha C V_{DD}^2 f$$

Short Circuit Current

- ❑ When transistors switch, both nMOS and pMOS networks may be momentarily ON at once
- ❑ Leads to a blip of “short circuit” current.
- ❑ $< 10\%$ of dynamic power if rise/fall times are comparable for input and output
- ❑ We will generally ignore this component

Power Dissipation Sources

- ❑ $P_{\text{total}} = P_{\text{dynamic}} + P_{\text{static}}$
- ❑ Dynamic power: $P_{\text{dynamic}} = P_{\text{switching}} + P_{\text{shortcircuit}}$
 - Switching load capacitances
 - Short-circuit current
- ❑ Static power: $P_{\text{static}} = (I_{\text{sub}} + I_{\text{gate}} + I_{\text{junct}} + I_{\text{contention}})V_{\text{DD}}$
 - Subthreshold leakage
 - Gate leakage
 - Junction leakage
 - Contention current

Dynamic Power Example

- ❑ 1 billion transistor chip
 - 50M logic transistors
 - Average width: 12λ
 - Activity factor = 0.1
 - 950M memory transistors
 - Average width: 4λ
 - Activity factor = 0.02
 - 1.0 V 65 nm process
 - $C = 1 \text{ fF}/\mu\text{m}$ (gate) + $0.8 \text{ fF}/\mu\text{m}$ (diffusion)
- ❑ Estimate dynamic power consumption @ 1 GHz.
Neglect wire capacitance and short-circuit current.

Solution

$$C_{\text{logic}} = (50 \times 10^6)(12\lambda)(0.025\mu\text{m} / \lambda)(1.8 \text{ fF} / \mu\text{m}) = 27 \text{ nF}$$

$$C_{\text{mem}} = (950 \times 10^6)(4\lambda)(0.025\mu\text{m} / \lambda)(1.8 \text{ fF} / \mu\text{m}) = 171 \text{ nF}$$

$$P_{\text{dynamic}} = [0.1C_{\text{logic}} + 0.02C_{\text{mem}}](1.0)^2(1.0 \text{ GHz}) = 6.1 \text{ W}$$

Dynamic Power Reduction

□ $P_{\text{switching}} = \alpha C V_{DD}^2 f$

□ Try to minimize:

- Activity factor
- Capacitance
- Supply voltage
- Frequency

Activity Factor Estimation

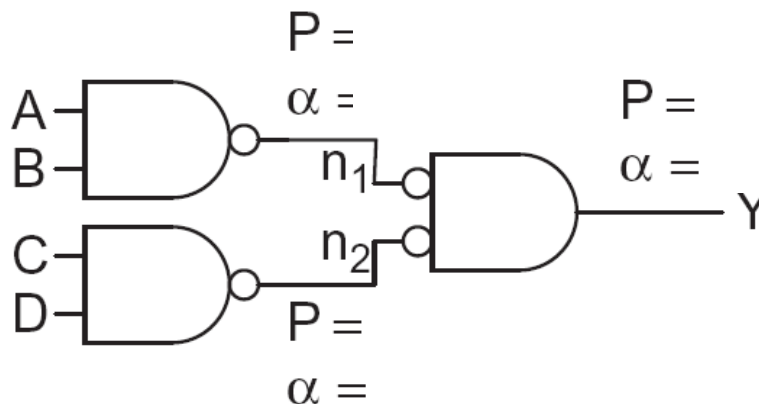
- ❑ Let $P_i = \text{Prob}(\text{node } i = 1)$
 - $\bar{P}_i = 1 - P_i$
- ❑ $\alpha_i = \bar{P}_i * P_i$
- ❑ Completely random data has $P = 0.5$ and $\alpha = 0.25$
- ❑ Data is often not completely random
 - e.g. upper bits of 64-bit words representing bank account balances are usually 0
- ❑ Data propagating through ANDs and ORs has lower activity factor
 - Depends on design, but typically $\alpha \approx 0.1$

Switching Probability

Gate	P_Y
AND2	$P_A P_B$
AND3	$P_A P_B P_C$
OR2	$1 - \bar{P}_A \bar{P}_B$
NAND2	$1 - P_A P_B$
NOR2	$\bar{P}_A \bar{P}_B$
XOR2	$P_A \bar{P}_B + \bar{P}_A P_B$

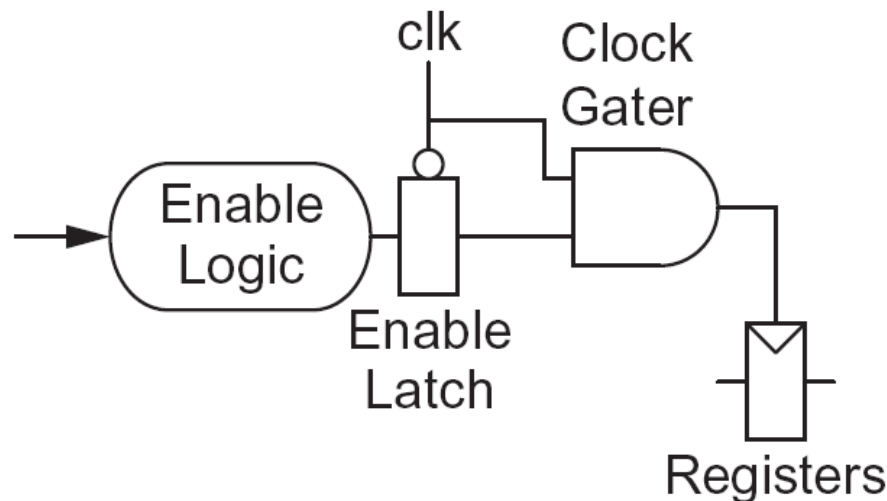
Example

- ❑ A 4-input AND is built out of two levels of gates
- ❑ Estimate the activity factor at each node if the inputs have $P = 0.5$



Clock Gating

- ❑ The best way to reduce the activity is to turn off the clock to registers in unused blocks
 - Saves clock activity ($\alpha = 1$)
 - Eliminates all switching activity in the block
 - Requires determining if block will be used

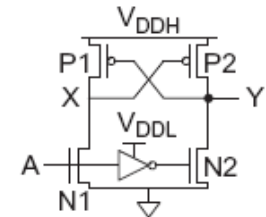


Capacitance

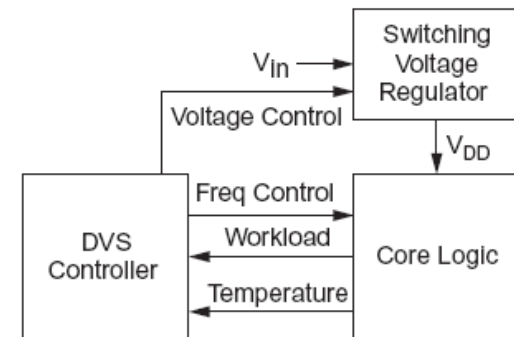
- ❑ Gate capacitance
 - Fewer stages of logic
 - Small gate sizes
- ❑ Wire capacitance
 - Good floorplanning to keep communicating blocks close to each other
 - Drive long wires with inverters or buffers rather than complex gates

Voltage / Frequency

- ❑ Run each block at the lowest possible voltage and frequency that meets performance requirements
- ❑ Voltage Domains
 - Provide separate supplies to different blocks
 - Level converters required when crossing from low to high V_{DD} domains



- ❑ Dynamic Voltage Scaling
 - Adjust V_{DD} and f according to workload



Static Power

- ❑ Static power is consumed even when chip is quiescent.
 - Leakage draws power from nominally OFF devices
 - Ratioed circuits burn power in fight between ON transistors

Static Power Example

- ❑ Revisit power estimation for 1 billion transistor chip
- ❑ Estimate static power consumption
 - Subthreshold leakage
 - Normal V_t : 100 nA/ μm
 - High V_t : 10 nA/ μm
 - High V_t used in all memories and in 95% of logic gates
 - Gate leakage 5 nA/ μm
 - Junction leakage negligible

Solution

$$W_{\text{normal-}V_t} = (50 \times 10^6)(12\lambda)(0.025\mu\text{m} / \lambda)(0.05) = 0.75 \times 10^6 \mu\text{m}$$

$$W_{\text{high-}V_t} = \left[(50 \times 10^6)(12\lambda)(0.95) + (950 \times 10^6)(4\lambda) \right] (0.025\mu\text{m} / \lambda) = 109.25 \times 10^6 \mu\text{m}$$

$$I_{\text{sub}} = \left[W_{\text{normal-}V_t} \times 100 \text{ nA}/\mu\text{m} + W_{\text{high-}V_t} \times 10 \text{ nA}/\mu\text{m} \right] / 2 = 584 \text{ mA}$$

$$I_{\text{gate}} = \left[(W_{\text{normal-}V_t} + W_{\text{high-}V_t}) \times 5 \text{ nA}/\mu\text{m} \right] / 2 = 275 \text{ mA}$$

$$P_{\text{static}} = (584 \text{ mA} + 275 \text{ mA})(1.0 \text{ V}) = 859 \text{ mW}$$

Subthreshold Leakage

- For $V_{ds} > 50 \text{ mV}$

$$I_{sub} \approx I_{off} 10^{\frac{V_{gs} + \eta(V_{ds} - V_{DD}) - k_{\gamma} V_{sb}}{S}}$$

- I_{off} = leakage at $V_{gs} = 0$, $V_{ds} = V_{DD}$

Typical values in 65 nm

$$I_{off} = 100 \text{ nA}/\mu\text{m} @ V_t = 0.3 \text{ V}$$

$$I_{off} = 10 \text{ nA}/\mu\text{m} @ V_t = 0.4 \text{ V}$$

$$I_{off} = 1 \text{ nA}/\mu\text{m} @ V_t = 0.5 \text{ V}$$

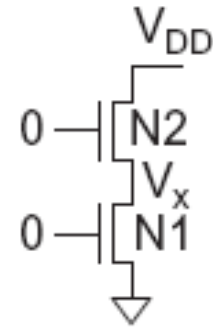
$$\eta = 0.1$$

$$k_{\gamma} = 0.1$$

$$S = 100 \text{ mV/decade}$$

Stack Effect

- ❑ Series OFF transistors have less leakage
 - $V_x > 0$, so N2 has negative V_{gs}



$$I_{sub} = \underbrace{I_{off} 10^{\frac{\eta(V_x - V_{DD})}{S}}}_{N2} = \underbrace{I_{off} 10^{\frac{-V_x + \eta((V_{DD} - V_x) - V_{DD}) - k_\gamma V_x}{S}}}_{N1}$$

$$V_x = \frac{\eta V_{DD}}{1 + 2\eta + k_\gamma}$$

$$I_{sub} = I_{off} 10^{\frac{-\eta V_{DD} \left(\frac{1 + \eta + k_\gamma}{1 + 2\eta + k_\gamma} \right)}{S}} \approx I_{off} 10^{\frac{-\eta V_{DD}}{S}}$$

- Leakage through 2-stack reduces ~10x
- Leakage through 3-stack reduces further

Leakage Control

- ❑ Leakage and delay trade off
 - Aim for low leakage in sleep and low delay in active mode
- ❑ To reduce leakage:
 - Increase V_t : *multiple V_t*
 - Use low V_t only in critical circuits
 - Increase V_s : *stack effect*
 - *Input vector control* in sleep
 - Decrease V_b
 - *Reverse body bias* in sleep
 - Or forward body bias in active mode

Gate Leakage

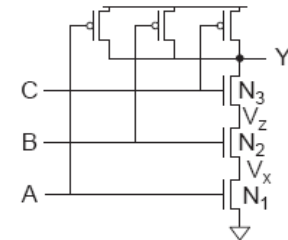
- ❑ Extremely strong function of t_{ox} and V_{gs}
 - Negligible for older processes
 - Approaches subthreshold leakage at 65 nm and below in some processes
- ❑ An order of magnitude less for pMOS than nMOS
- ❑ Control leakage in the process using $t_{ox} > 10.5 \text{ \AA}$
 - High-k gate dielectrics help
 - Some processes provide multiple t_{ox}
 - e.g. thicker oxide for 3.3 V I/O transistors
- ❑ Control leakage in circuits by limiting V_{DD}

NAND3 Leakage Example

□ 100 nm process

$$I_{gn} = 6.3 \text{ nA} \quad I_{gp} = 0$$

$$I_{offn} = 5.63 \text{ nA} \quad I_{offp} = 9.3 \text{ nA}$$



Input State (ABC)	I_{sub}	I_{gate}	I_{total}	V_x	V_z
000	0.4	0	0.4	stack effect	stack effect
001	0.7	0	0.7	stack effect	$V_{DD} - V_t$
010	0	1.3	1.3	intermediate	intermediate
011	3.8	0	10.1	$V_{DD} - V_t$	$V_{DD} - V_t$
100	0.7	6.3	7.0	0	stack effect
101	3.8	6.3	10.1	0	$V_{DD} - V_t$
110	5.6	12.6	18.2	0	0
111	28	18.9	46.9	0	0

Data from [Lee03]

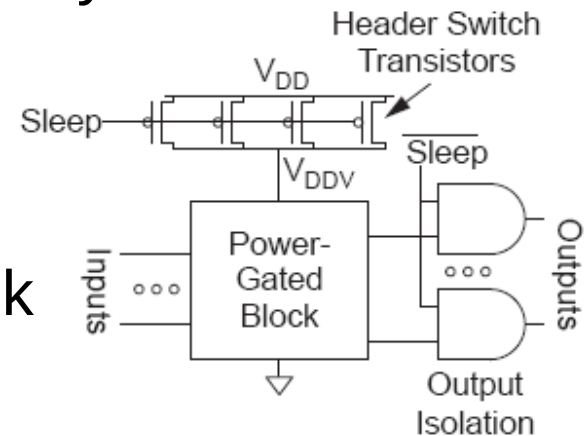
Junction Leakage

- ❑ From reverse-biased p-n junctions
 - Between diffusion and substrate or well
- ❑ Ordinary diode leakage is negligible
- ❑ Band-to-band tunneling (BTBT) can be significant
 - Especially in high- V_t transistors where other leakage is small
 - Worst at $V_{db} = V_{DD}$
- ❑ Gate-induced drain leakage (GIDL) exacerbates
 - Worst for $V_{gd} = -V_{DD}$ (or more negative)

Power Gating

- ❑ Turn OFF power to blocks when they are idle to save leakage

- Use virtual V_{DD} (V_{DDV})
- Gate outputs to prevent invalid logic levels to next block



- ❑ Voltage drop across sleep transistor degrades performance during normal operation
 - Size the transistor wide enough to minimize impact
- ❑ Switching wide sleep transistor costs dynamic power
 - Only justified when circuit sleeps long enough